

GrounUp Enterprise Specification

Phase 06 - AI Construction Intelligence Platform

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Executive Summary

Phase 06 defines the governed AI brain of GrounUp: coordinated specialist agents, tenant-isolated knowledge, multimodal document intelligence, human approvals, explainability, evaluation, white labeling, and marketplace extensibility.

Specialist Agents

ID	Agent	Purpose
AI-AGT-001	Orchestration Agent	Routes tasks, decomposes work, coordinates specialist agents, and assembles final responses.
AI-AGT-002	Document Intelligence Agent	Reads plans, specifications, addenda, reports, contracts, manuals, and project files.
AI-AGT-003	Takeoff Agent	Extracts measurable quantities and generates traceable takeoff records.
AI-AGT-004	Estimating Agent	Creates estimate-ready line items, resource buildups, pricing scenarios, and assumptions.
AI-AGT-005	Production Agent	Recommends crews, equipment spreads, production rates, durations, and fuel usage.
AI-AGT-006	Risk Agent	Identifies scope gaps, conflicts, uncertainty, constructability risk, and commercial exposure.
AI-AGT-007	Scheduling Agent	Generates sequencing, durations, constraints, and schedule-impact forecasts.
AI-AGT-008	Project Management Agent	Supports RFIs, submittals, change events, budget forecasting, and progress controls.
AI-AGT-009	Field Operations Agent	Analyzes daily reports, field production, photos, timecards, and job conditions.
AI-AGT-010	Safety Agent	Supports JHAs/JSAs, hazard identification, training checks, and safety observations.
AI-AGT-011	Fleet Agent	Optimizes assignments, utilization, maintenance forecasts, rental decisions, and fuel.
AI-AGT-012	Procurement Agent	Compares quotes, lead times, vendor performance, substitutions, and buyout status.
AI-AGT-013	CRM Agent	Scores leads, drafts communications, recommends follow-ups, and forecasts pipeline.
AI-AGT-014	HR and Workforce Agent	Supports staffing, certification checks, training, attendance, and workforce planning.
AI-AGT-015	Financial Intelligence Agent	Forecasts cash flow, profitability, cost-to-complete, and variance drivers.
AI-AGT-016	Executive Agent	Produces executive summaries, KPI explanations, strategic alerts, and decision briefs.
AI-AGT-017	Quality Agent	Supports inspections, punch lists, nonconformance review, and corrective actions.
AI-AGT-018	Compliance Agent	Checks tenant policies, contract requirements, records retention, and regulatory controls.
AI-AGT-019	Customer Portal Agent	Answers authorized customer questions using project-scoped, approved information.
AI-AGT-020	Vendor and Subcontractor Agent	Supports onboarding, compliance, bid leveling, performance, and communication.

AI Engines

AI-ENG-001 - Agent Orchestration Engine
AI-ENG-002 - Intent and Task Classification Engine
AI-ENG-003 - Context Assembly Engine
AI-ENG-004 - Retrieval and Reranking Engine
AI-ENG-005 - Construction Knowledge Graph Engine
AI-ENG-006 - Prompt and Policy Resolution Engine
AI-ENG-007 - Tool and API Execution Engine
AI-ENG-008 - Human Approval Engine
AI-ENG-009 - Confidence and Uncertainty Engine
AI-ENG-010 - Citation and Provenance Engine
AI-ENG-011 - Memory Management Engine
AI-ENG-012 - Model Routing Engine
AI-ENG-013 - Cost and Token Control Engine
AI-ENG-014 - AI Safety and Guardrail Engine
AI-ENG-015 - Evaluation and Benchmark Engine
AI-ENG-016 - Feedback and Learning Engine
AI-ENG-017 - Multimodal Processing Engine
AI-ENG-018 - Tenant Knowledge Isolation Engine
AI-ENG-019 - AI Audit and Observability Engine
AI-ENG-020 - AI Marketplace Packaging Engine

Business Rules

AI-BR-001 - Material decisions require human approval

AI outputs may not autonomously commit estimates, contracts, purchase orders, payroll, safety closures, or customer-facing commitments unless an approved automation policy explicitly permits it.

AI-BR-002 - Source-grounded answers

Project-specific factual answers must cite authorized source records or be labeled as assumptions.

AI-BR-003 - Tenant isolation

AI retrieval, memory, logs, embeddings, and model context must remain isolated by tenant and authorized scope.

AI-BR-004 - No silent writeback

AI may propose changes but may not alter governed master data without an auditable approval workflow.

AI-BR-005 - Confidence disclosure

Every recommendation affecting cost, quantity, duration, safety, compliance, or risk must display confidence and uncertainty.

AI-BR-006 - Prompt versioning

Production prompts must be versioned, tested, approved, and rollback-capable.

AI-BR-007 - Model independence

Business logic and canonical data must not be embedded exclusively in one model provider.

AI-BR-008 - Least privilege tools

Agents receive only the tools and data permissions required for their assigned task.

AI-BR-009 - Retention governance

Conversation, prompt, embedding, audit, and feedback retention follow tenant policy and legal holds.

AI-BR-010 - White-label disclosure control

Tenant branding may customize the AI identity while retaining required legal and safety disclosures.

AI-BR-011 - High-risk escalation

Low-confidence or conflicting results must be escalated instead of presented as definitive.

AI-BR-012 - Calculation traceability

Numeric outputs must retain inputs, formulas, units, conversions, source rates, and manual adjustments.

AI-BR-013 - Feedback separation

User feedback must not update production behavior until validated through the learning pipeline.

AI-BR-014 - Marketplace review

Third-party agents must pass security, privacy, functional, and compatibility review before publication.

AI-BR-015 - Data minimization

Only the minimum required data may be placed into model context or external model calls.

AI-BR-016 - Sensitive-data controls

Sensitive HR, financial, legal, medical, and identity information requires enhanced access and masking controls.

AI-BR-017 - Action confirmation

Destructive, external, financial, contractual, or irreversible actions require explicit confirmation.

AI-BR-018 - Evaluation gate

No agent or prompt version may be promoted without passing its required evaluation suite.

AI-BR-019 - Incident containment

AI incidents must support rapid disablement by agent, prompt, tool, model, tenant, or feature.

AI-BR-020 - Canonical writeback

Approved AI outputs write back only through canonical APIs and governed entity services.

Validation Rules

AI-VR-001 - Agent ID uniqueness

Agent identifiers must be globally unique within the registry.

AI-VR-002 - Prompt checksum

Every prompt version must store an immutable checksum.

AI-VR-003 - Citation validity

Citations must resolve to an authorized source and stable locator.

AI-VR-004 - Confidence range

Confidence values must be between 0 and 1.

AI-VR-005 - Approval identity

Approvals must capture actor, timestamp, decision, scope, and version.

AI-VR-006 - Tool allowlist

Tool calls must match the agent deployment allowlist.

AI-VR-007 - Tenant scope

Every AI record must include tenant scope unless explicitly global.

AI-VR-008 - Model status

Only approved active model profiles may serve production traffic.

AI-VR-009 - Evaluation completeness

Required evaluation cases must complete before promotion.

AI-VR-010 - Retention date

Retention or deletion dates must be computable for governed records.

AI-VR-011 - Unit consistency

Quantitative AI outputs must include canonical unit codes.

AI-VR-012 - Schema conformity

Structured outputs must validate against the assigned JSON schema.

AI-VR-013 - Memory sensitivity

Sensitive content cannot be stored in long-term memory without policy authorization.

AI-VR-014 - Version dependency

Agent deployments must pin prompt, policy, model, and tool versions.

AI-VR-015 - Rollback target

A production deployment must identify a valid rollback target.

AI-VR-016 - Marketplace signature

Published packages must be signed and checksum verified.

AI-VR-017 - Cost limit

Requests exceeding tenant cost limits must be rejected or routed for approval.

AI-VR-018 - PII masking

Configured sensitive fields must be masked before external model transmission.

AI-VR-019 - Incident severity

AI incidents must use an approved severity value.

AI-VR-020 - Audit immutability

AI audit events must be append-only and tamper-evident.

Controlled Requirements

[illegible]

[illegible]

ID	Area	Requirement
AI-REQ-106	Multimodal	The platform shall provide governed multimodal capability 106 with tenant isolation, auditability, role-based authorization, version control, and human-review controls wh
AI-REQ-107	Prompts	The platform shall provide governed prompts capability 107 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where
AI-REQ-108	Models	The platform shall provide governed models capability 108 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where
AI-REQ-109	Human Oversight	The platform shall provide governed human oversight capability 109 with tenant isolation, auditability, role-based authorization, version control, and human-review contr
AI-REQ-110	Explainability	The platform shall provide governed explainability capability 110 with tenant isolation, auditability, role-based authorization, version control, and human-review controls v
AI-REQ-111	Security	The platform shall provide governed security capability 111 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where
AI-REQ-112	Privacy	The platform shall provide governed privacy capability 112 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where
AI-REQ-113	Tenant Isolation	The platform shall provide governed tenant isolation capability 113 with tenant isolation, auditability, role-based authorization, version control, and human-review control
AI-REQ-114	Observability	The platform shall provide governed observability capability 114 with tenant isolation, auditability, role-based authorization, version control, and human-review controls v
AI-REQ-115	Evaluation	The platform shall provide governed evaluation capability 115 with tenant isolation, auditability, role-based authorization, version control, and human-review controls wh
AI-REQ-116	Learning	The platform shall provide governed learning capability 116 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where
AI-REQ-117	Marketplace	The platform shall provide governed marketplace capability 117 with tenant isolation, auditability, role-based authorization, version control, and human-review controls w
AI-REQ-118	White Label	The platform shall provide governed white label capability 118 with tenant isolation, auditability, role-based authorization, version control, and human-review controls wh
AI-REQ-119	API	The platform shall provide governed api capability 119 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where mat
AI-REQ-120	UI/UX	The platform shall provide governed ui/ux capability 120 with tenant isolation, auditability, role-based authorization, version control, and human-review controls where m